Aly Dhedhi

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EDUCATION

Jumeirah College
High School Diploma **Dubai, UAE**
August 2024–June 2026

- Weighted GPA ~ 4.1/ 4.0, Unweighted GPA ~ 4.0/4.0
- Pursuing GCSE qualifications in core subjects (Math, English Literature and Language, Sciences) along with: Economics, Geography, Food Science and Design Technology.
- Involved in: MUN, WSC, JC Econ magazine; Founder of Mu Alpha Theta's JC chapter
- Captain of JC Rugby Squad (2024-25)

WORK & LEADERSHIP EXPERIENCE

JEPA for Health AI — Working Group **Cambridge,MA,USA**
(MIT Critical Data) May 2026–Present

- Participating in the JEPA for Health AI working group led by Leo Anthony Celi at MIT Critical Data .
- Exploring Joint Embedding Predictive Architectures (JEPA) — a self-supervised framework predicting in latent representation space to learn abstract, high-level clinical concepts rather than reconstructing raw inputs.
- Contributing to research threads including multimodal clinical discordance detection (chest X-rays, lab values, clinical notes), bedside alarm precision via temporal waveform abstraction, and causal patient trajectory modeling.

Clinical Bias Research in Obesity Clinical Guidelines **Cambridge, MA,USA**
(MIT Critical Data) March 2026–Present

- Leading a research pipeline identifying geographic and demographic bias in clinical literature through citation extraction, representation analysis, and structured dataset generation.
- Extracting and processing citation networks from the AACE/ACE 2016 Clinical Practice Guidelines for Obesity (DOI: 10.4158/EP161365.GL).
- Collaborating with researchers from UCLA, University of Pennsylvania (UPENN), and National University of Singapore (NUS) within the MIT Critical Data global network.
- Colab: Obesity Guidelines Bias Pipeline

Hospital-Invariant ICU Representation Learning & Counterfactual Modeling **Dubai, UAE**
(SRG) March 2026–April 2026

- Architected an end-to-end ICU mortality prediction pipeline using MIMIC-IV, processing high-dimensional tabular features from clinical time-series and labs.
- Evaluated Logistic Regression, Random Forest, XGBoost, and LightGBM under class imbalance, achieving AUROC > 0.84 and AUPRC ≈ 0.28.
- Used UMAP for low-dimensional representation and validation of latent patient structure. Built a modular, reproducible pipeline spanning data extraction through interpretability.
- Paper submitted to IEEE CIBCB 2026

- Paper(As of now it is in Preprint): doi.org/10.21203/rs.3.rs-9612331/v1
- Colab: Unsupervised Cluster-Augmented Mortality Prediction in MIMIC-IV Using Tabular ICU T…

Computational Peptide Design for BBB-Penetrant EGFRvIII Targeting **Dubai, UAE**
(YRI) November 2025–March 2026

- Architected a cloud-native in silico pipeline for BBB-penetrant peptide design targeting EGFRvIII in glioblastoma.
- Designed four Angiopep-2/GE11 fusion constructs with structural variations (cyclization, terminal capping, retro-enantio) to probe stability and binding geometry.
- Generated de novo peptide structures via ESMFold API and conducted geometric docking pre-screening using PatchDock.
- Identified a cyclized lead candidate with highest predicted interface interaction (≈281 hydrophobic contacts, 33 salt bridges) and BBB likelihood.
- Built a fully reproducible Google Colab pipeline using open-source tools, enabling end-to-end execution without local dependencies.
- Authored and submitted research paper to IEEE CIBCB 2026.
- Paper:
- Colab: BBB-Penetrant

Massachusetts Institute of Technology (MIT) **Cambridge, MA, USA**
Research & AI Developer November 2025 –Present

- Leverage AI and representation learning to model high-dimensional clinical data, constructing unified patient embeddings and longitudinal trajectory representations from MIMIC-IV.
- Design and implement a two-stage pipeline combining multimodal embedding (scaled labs + diagnosis text embeddings) with clustering (KMeans) to derive latent clinical states and patient-level temporal trajectories.
- Engineer trajectory-level features (state transitions, path length, temporal duration, cluster displacement) to quantify disease progression dynamics and enable secondary clustering of patient progression patterns.
- Analyze and visualize large-scale patient transitions (~millions of temporal steps) through embedding spaces, identifying structurally distinct progression phenotypes across cohorts.
- Validate structure and robustness using dimensionality reduction (PCA/UMAP), cluster stability metrics, and transition pattern analysis, producing reproducible artifacts for downstream modeling.
- Collaborate with Manolis Kellis and an interdisciplinary team to translate trajectory-based representations into scalable, hospital-invariant biomedical learning systems.

The Citizens Foundation(TCF) **Dubai, UAE**
Student Fundraiser and Volunteer July 2022 – Present

- Raised over 23,000$ to support underprivileged children's education; equivalent to a year's worth of education for 166 children
- Invited to deliver speeches at national fundraising events to advocate for education access and encourage donor participation
- Coordinated fundraising campaigns and engaged peers to maximize outreach and contributions; Recognised at national dinners as the top fundraiser for the youth cohort twice (December 2023, December 2025)

HCA Florida – West Palm Beach **West Palm Beach, FL**
Gastroenterologist Assistant / Clinical Shadowing Student *July 2025 – August 2025*

- Assisted a gastroenterologist in patient consultations, observing diagnostic workflows and treatment planning processes

- Gained hands-on exposure to clinical procedures, including endoscopic techniques, biopsy collection, and patient monitoring
- Developed understanding of gastroenterology practice, patient care, and interdisciplinary collaboration in a hospital setting

Boy Scouts of America, Troop 813 **Dubai, UAE**
Troop Guide/ Patrol Leader *August 2021 – Present*

- Current Star Scout; Eagle Scout expected by mid 2027
- Led patrols, organized community service projects, and mentored younger scouts in leadership and civic responsibility
- Key merit badges earned: Lifesaving, First Aid, Shotgun Shooting, Rifle Shooting, Citizenship in the Nation, World, and Community

ADDITIONAL SKILLS AND INTERESTS

Licenses and Certifications: CITI Data or Specimens Only Research; CITI Conflict of Interest; Trinity College London Piano (Grade 5)
Technical:
Python (intermediate; scientific computing, data pipelines), PyTorch (foundational), scikit-learn, NumPy, pandas, Matplotlib; BioPython (sequence analysis, physicochemical profiling); protein structure tools/APIs (ESMFold); molecular docking workflows (PatchDock); Google Colab (cloud-native pipelines); HTML5 (fluent); Figma (UI/UX design, prototyping); Microsoft Office Suite, Google Workspace; experience with Manus, MGX, Bolt AI, Lovable, Base44

Laboratory & Research:
Computational molecular biology and peptide design; in silico structural modeling and docking-based screening; physicochemical characterization and BBB permeability heuristics; experimental design and reproducible pipeline development; literature-driven hypothesis formulation and scientific writing; developed web-based platform for peptide research compilation

Languages: English (fluent), Urdu (fluent), Arabic (intermediate), Mandarin (intermediate)
Interests: Biomedical research, AI in healthcare, clinical phenotyping, oncology, Muay Thai, tennis, golf, fundraising initiatives